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ON THE GENUS *NOSTOCHOPSIS* WOOD

By

T. V. DESIKACHARY

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ON THE GENUS *NOSTOCHOPSIS* WOOD

BY T. V. DESIKACHARY

(With thirty-eight Figures in the Text)

THE genus *Nostochopsis* was established by Wood in 1869 on the type species *N. lobatus* which was collected by him in Philadelphia. Hansgirg, in 1893, described a new variety of *N. lobatus*, viz., var. *stagnalis* from Bohemia. Schmidle (1900 a) recorded a second species, *N. rupestris*, from Bombay, which he later (1900 b) renamed as *N. Hansgirgii*. A third species *N. Wichmannii* was described from Indo-Malayan Archipelago by Mme. Weber van Bosse (1913). Gardner (1927) recorded from China a new variety of *N. Hansgirgii*, viz., var. *sphaericus*. Geitler (1932) in his *Cyanophyceae* recognized three species and two varieties, viz., *N. lobatus*, *N. lobatus* var. *stagnalis*, *N. Hansgirgii*, *N. Hansgirgii* var. *sphaericus* and *N. Wichmannii*. But, later on he (Geitler, 1933 and Geitler and Ruttner, 1935-36) from a study of some material of *N. lobatus* from Sunda Islands came to the conclusion that *N. Wichmannii* and *N. Hansgirgii* var. *sphaericus* should be included under *N. lobatus* Wood and emended the description of *N. lobatus* Wood accordingly. He kept *N. Hansgirgii* as a species distinct from *N. lobatus*. As regards *N. lobatus* var. *stagnalis*, he expressed the view that it was very doubtful whether it was a *Nostochopsis* at all.

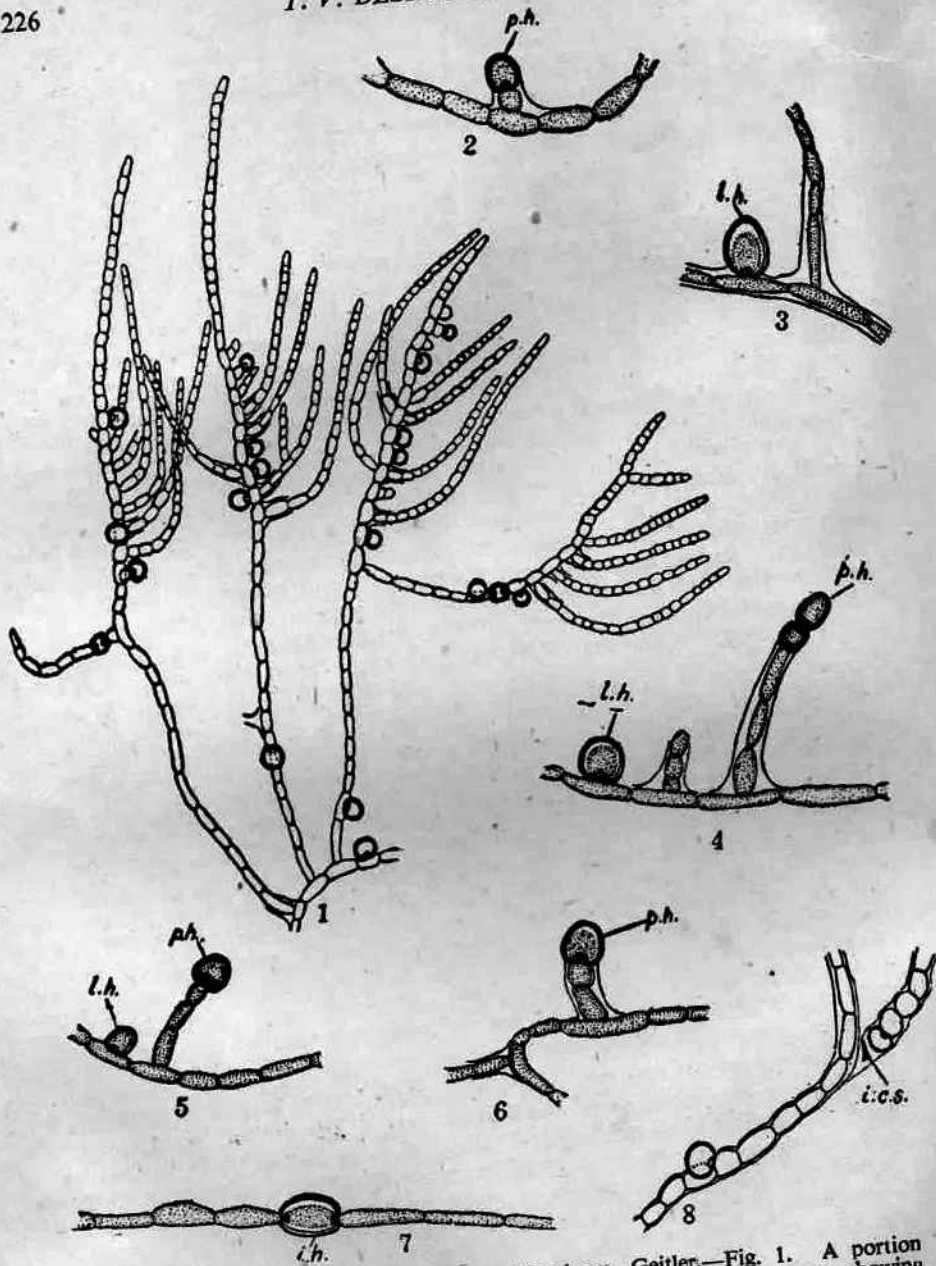
In 1934, Bharadwaja described a new species *Nostochopsis radians* from the Jog Falls region in South India.

In the same year Frémy and Feldmann (1934) made a detailed study of a collection of *N. lobatus* from Banyuls in France and came to the conclusion that all the previously recorded species of *Nostochopsis* should be included under *N. lobatus*. In other words, not only did they agree with Geitler (1933) regarding the inclusion of *N. Wichmannii* and *N. Hansgirgii* var. *sphaericus* but also went further and suggested that *N. Hansgirgii* and *N. radians* also should be included under *N. lobatus*. As regards *N. lobatus* var. *stagnalis* they also like Geitler (Geitler, 1933 and Geitler and Ruttner, 1935-36) doubted whether it was a *Nostochopsis* at all.

Prof. M. O. P. Iyengar very kindly placed at the writer's disposal material of two species of *Nostochopsis*, viz., *N. lobatus* and *N. radians*, which he collected from South India and suggested that a detailed study of these two species should be made by the writer with a view to find out how far the writer's observations agreed with or differed from those of the previous workers and incidentally also to increase our knowledge of these two species, if possible.

Nostochopsis lobatus

This alga was collected from two places in South India, viz., (1) near top-slips in the Anamalais and (2) in a stream at Tirumoorthy



Text-figs. 1-8. *Nostochopsis lobatus* Wood em. Geitler.—Fig. 1. A portion of the alga showing its habit. $\times 300$. Figs. 2-5. Portions of filaments showing sessile and pedicellate lateral heterocysts. $\times 750$. Fig. 6. Portion of a filament with a pedicellate lateral heterocyst. $\times 750$. Fig. 7. Portion of a filament with a false-intercalary heterocyst. $\times 750$. Fig. 8. Portion of a filament with a false-intercalary heterocyst. $\times 750$. (l.h. = sessile lateral heterocyst; p.h. = pedicellate lateral heterocyst; i.h. = intercalary heterocyst; i.c.s. = discs of intercellular substance.

Hills near Coimbatore. In both these places it was found growing attached to rocks. The following description is based on the material collected from the Anamalais, as a good quantity of this material was available in the collection. The alga collected from the Tirumoorthy Hills agreed with the Anamalais form in all respects.

The thallus is hollow and globular and attains a diameter of about $1-1\frac{1}{2}$ inches. It consists of an inner portion of entangled filaments forming the inner lining of the hollow thallus. From these erect branches numerous branches are repeatedly formed and these also bend upwards and run in a radial direction (Text-fig. 1). The filaments of the inner portion are narrow and made up of cylindrical cells $1.3-2.6\mu$ broad and $9.2-14.5\mu$ long. The radially disposed filaments taper somewhat towards their apex. They are torulose below and are only slightly constricted at the cross-walls in the apical portions. The terminal cell is conical. The filaments consist of short or long barrel-shaped cells which are $3.9-5.2$ (-6.6) μ broad and $6.5-9.2\mu$ long. In the terminal portions the cells are about 2μ broad.

The filaments are provided with a very thin closely investing sheath. The sheath is very indistinct and can be seen well only at the base of the branches. The filaments lie embedded in a common amorphous gelatinous matrix.

Both lateral and intercalary heterocysts are present. The lateral heterocysts are very common and may be either sessile or pedicellate on a stalk consisting of 1-3 cells (Text-figs. 2-6). Intercalary heterocysts are less common (Text-fig. 7). In many cases the trichome gets broken off by the side of an intercalary heterocyst which then looks like a terminal heterocyst. But its intercalary nature could easily be recognized by its being two-pored and often by the presence of a portion of the sheath left by the broken filament. The lateral heterocysts are ovate to obovate and measure $5.2-7.9\mu$ broad and are as long as or slightly longer than broad. The intercalary heterocysts are ellipsoidal to cylindrical and are more or less of the same breadth as the lateral ones but are usually longer.

All the branches are true. The branches first arise as a small protuberance in the middle of an ordinary cell. Later this protuberance is cut off as a lateral cell. This lateral cell by further growth develops into a new branch. During its growth it pierces through the parent sheath and later secretes its own sheath. The limits of the branch sheath and that of the parent sheath are indistinct. Branches given off from the erect portions usually arise on one side of the filament or sometimes on both sides of the filament. False branching occurs very occasionally (Text-fig. 8).

Hormogones.—The formation of hormogones and their germination have been previously recorded by Geitler and Ruttner (1935-36) in their Sunda Island material of *N. lobatus*. Only a few stages of the germination of the hormogones were observed by them (Text-figs. 9-11). The writer was able to observe numerous cases of germination



Text-figs. 9-23. *Nostochopsis lobatus* Wood em. Geitler.—Figs. 9-11. Hormogone formation and their germination (after Geitler and Ruttner, 1935-36).
 Fig. 12. Apical portion of a filament showing hormogonial branches. $\times 600$.
 Fig. 13. Apical portion of a filament showing a hormogone that has been cut off by discs of intercellular substance. $\times 600$. Figs. 14-23. Stages in the germination of the hormogones. (Figs. 14-16, $\times 600$; 18, 19, 21, $\times 600$; 17, 20, 22, 23, $\times 400$.)

of the hormogones in the Anamalais material. All the stages of germination right up to the formation of a typical young branched *Nostochopsis* filament were observed in the material.

The hormogones in the Anamalais material are formed at the ends of short branches (Text-fig. 12). The branches giving rise to hormogones are not attenuated at the terminal portion as in the ordinary trichome but are of the same thickness throughout. The cells of the end portions of these branches become more or less rounded and this end portion is finally cut off as a hormogone by the formation of discs of intercellular substance (Text-fig. 13). The hormogones consist of 8-20 cells. The cells are spherical to sub-spherical and are 4-5.2 μ broad. The cells of the hormogone after liberation soon begin to divide. The terminal cell at one end of the hormogone becomes a heterocyst. The cells close to the heterocyst remain more or less sub-spherical to spherical but the cells further away elongate and become barrel-shaped. The cells at the extreme end become longer and narrower (Text-figs. 14, 15). The trichome soon secretes a thin hyaline sheath. This sheath could, however, be seen distinctly at the apex only. The hormogone elongates further and all the cells become narrower and barrel-shaped or cylindrical. Branches and lateral heterocysts are formed later on and ultimately a typical young branched *Nostochopsis* filament is formed (Text-figs. 17, 19, 20). Sometimes during the germination of the hormogone, instead of a single heterocyst being formed at one end only, a heterocyst may be formed at both the ends (Text-fig. 18). Very occasionally a single intercalary heterocyst only is formed and both the ends grow freely and contribute to the length of the filament (Text-fig. 21). In some cases no heterocysts are formed either at the end or in an intercalary position and the hormogone merely elongates and develops into a long narrow trichome, which later forms lateral heterocysts and branches (Text-figs. 22, 23).

No spores were observed in the alga.

This alga agrees in all respects with *N. lobatus* Wood emend. Geitler. It has a hollow thallus and possesses both lateral and intercalary heterocysts. But it differs from it in having the slightly smaller dimensions.

Nostochopsis radians

This species was described by Bharadwaja (1934) from material which had been collected by the late Prof. M. A. Sampathkumaran from a stream in the Jog Falls region in the Mysore State. A portion of the same original collection of this material was kindly placed at the writer's disposal by Prof. Sampathkumaran. The writer had at his disposal two later collections also of this species made by Prof. M. O. P. Iyengar, one from a stream in the same area near Jog Falls and another from a stream in Tenmalai in the Travancore State. The following account is based on a study of the materials collected by Prof. Sampathkumaran and by Prof. Iyengar from the Jog area. The Tenmalai form agrees very closely with the Jog form in all respects.

The alga forms a solid, spherical or hemispherical thallus and is olive-green in colour. The thallus is mucilaginous and is slimy to the touch.* The thallus consists of branched filaments arranged more or less radially. The basal portion consists of old contorted narrow filaments and these filaments give rise to erect branches. These branches are always true and their development agrees with those described by Bharadwaja (1934, p. 2). These branches also occupy a radial position. The filaments end in long narrow projecting threads (Text-figs. 24, 25). In the old portions of this alga the filaments show a thin firm sheath. In the younger portions of the alga, however, the sheath is not distinctly seen. The trichome is blue-green in colour. In the basal portion the cells are short and barrel-shaped and in the upper portion they are long and barrel-shaped. These cells are $3.9-7.9\mu$ broad and $3.9-13\mu$ long. In the terminal portions of the filaments the cells are narrower and nearly cylindrical and measure $1.3-2.6$ (-3.9) μ broad and $(6.6) 9.2-21\mu$ long.

The heterocysts are always lateral and are formed in the basal and the upper regions and are absent in the long terminal threads. The lateral heterocysts may be either sessile or pedicellate, the stalk in the latter case consisting of 1-3 cells (Text-figs. 24, 26-28). These heterocysts are ovate to obovate or ellipsoidal and are $5.2-10.5\mu$ broad and $5.2-11.5\mu$ long. In spite of very careful examination of a large amount of material, the writer was unable to find even a single instance of an intercalary heterocyst.

No hormones were found in the material.

The writer's observations on the alga generally agreed with those of Bharadwaja except for the fact that the thallus is mucilaginous and not non-mucilaginous as described by Bharadwaja.

DISCUSSION

According to Wood's original diagnosis (1872) the thallus in *N. lobatus* is hollow and the heterocysts are lateral, either sessile or pedicellate. Wille (1884) recorded intercalary heterocysts also in this species and later Hansgirg (1893) and others also recorded intercalary heterocysts in the alga. Geitler (1933) and Geitler and Ruttner (1935-36) considered that the nature of the thallus in *N. lobatus*, whether solid or hollow, is mainly due to the age of the alga and so should not be regarded as a distinguishing feature of the alga. Frémy and Feldmann (1934) actually found that the thallus of *N. lobatus* is solid when young and becomes hollow later on as previously suggested by Geitler. From this it is clear that the thallus in *N. lobatus* can be solid or hollow according to its age and possesses both lateral and intercalary heterocysts.

* It may be pointed out here that Bharadwaja (1934, p. 1) described this alga as non-mucilaginous. But the writer found that the alga was definitely mucilaginous. The mucilage seems to be of such a very thin consistency that it is easily dissolved away by long preservation in 4% Formalin. In a collection of this alga from Jog preserved in Form-alcobol, the thallus showed plenty of mucilage in it.

in these two forms come quite within the range of the dimensions recorded in *N. lobatus* by Geitler, Frémy and Feldmann. Hence the cell-dimensions cannot be used for distinguishing these two forms from *N. lobatus*. As regards (2), all the three forms are known to have lateral as well as intercalary heterocysts. This feature also cannot be considered as a point for separating these forms. As regards (3), Geitler considers that the solid or hollow nature of the thallus is mainly due to the age of alga and so should not be regarded as a feature for distinguishing these two forms from *N. lobatus*. Hence he included the two algæ *N. Wichmannii* W. v. Bosse and *N. Hansgirgii* Schmidle var. *sphaericus* Gardner under *N. lobatus* Wood and emended the diagnosis of the species accordingly.

The writer feels that Geitler is quite justified in including *N. Wichmannii* and *N. Hansgirgii* var. *sphaericus* under *N. lobatus* Wood emend. Geitler. Clearly these two forms must be considered as young plants of *N. lobatus*. Frémy and Feldmann (1934) agreed with Geitler's inclusion of the two forms under *N. lobatus* Wood emend. Geitler as quite valid. But they went further and suggested that *N. Hansgirgii* and *N. radians* also should be included under *N. lobatus*. The writer, however, is not prepared to accept this suggestion for the following reasons: *Nostochopsis Hansgirgii* (Text-figs. 29-33, 35-37) has a solid thallus. The filaments are less branched and very much narrower than those of *N. lobatus*. They measure (0.6-0.9-) 1-3 μ in breadth. Both lateral and intercalary heterocysts are present in this alga. Dixit (1936, p. 96) has recorded this alga for a second time. He described the alga as "growing on rocks in a rivulet near Goregaon in Bombay". The measurements given by Dixit agree very closely with those given by Schmidle. The dimensions of *N. Hansgirgii* are much smaller than those of *N. lobatus* and the nature of the branching is different. This is evidently a smaller species. The mature filament in this species is 1-3 μ broad, whereas in *N. lobatus* it is 4-5.5 (-9) μ . No transitional dimensions are seen in the alga, connecting it with *N. lobatus*. It is best, therefore, to keep this species separate until further data are available from a study of more material of *N. Hansgirgii*.

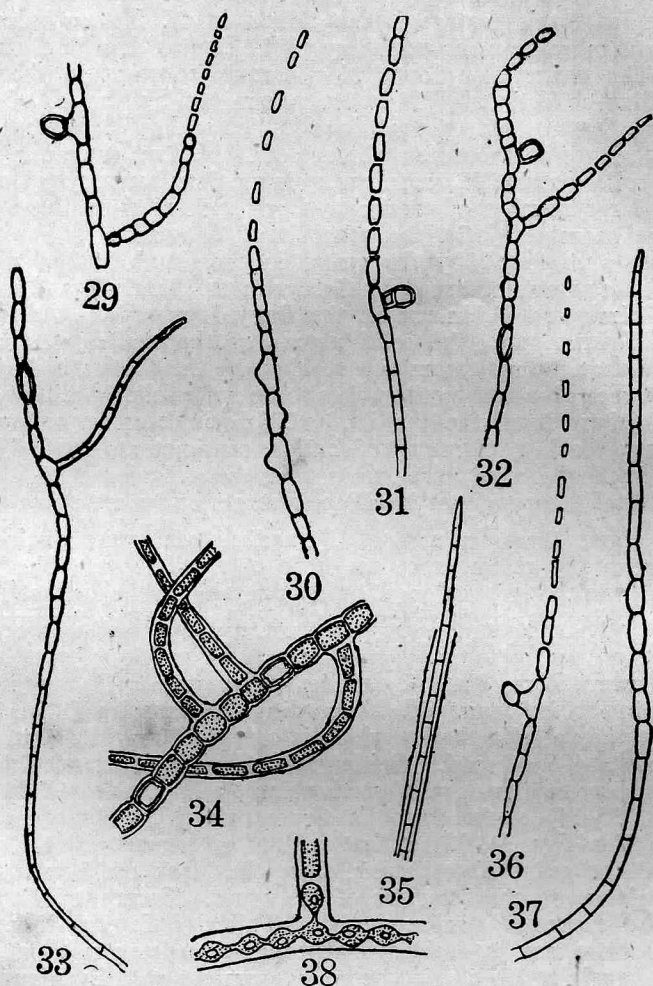
Nostochopsis radians as described by Bharadwaja differs from *N. lobatus* in the following points: (1) the unbranched character of the peripheral threads and the profuse branching in the middle region, (2) the absence of intercalary heterocysts, (3) the smaller and solid nature of the thallus, (4) the presence of a thick, hyaline and firm sheath, and (5) the non-mucilaginous nature of the thallus. The last two points of differences mentioned by Bharadwaja, viz., (1) the presence of a firm sheath in *N. radians* and (2) the absence of mucilage in the thallus need not be seriously taken into consideration since the presence of a thin hyaline sheath is now known to occur in *N. lobatus* also and the thallus of *N. radians* also is mucilaginous as has already been pointed out earlier by the writer. Only the first three points may now be considered. As regards the first point, viz., the unbranched nature of the peripheral threads and the profuse branching in the

middle region, this feature is not found in *N. lobatus* where the branching is more near the periphery of the thallus than in the middle region. The writer also found this feature constant in the three South Indian collections of *N. radians* examined by him. As regards the second feature, viz., the absence of intercalary heterocysts, this again appears to be a constant feature of *N. radians* since in none of the three collections of *N. radians* were intercalary heterocysts observed though a large number of specimens from each of these three collections were examined. On the other hand, a number of intercalary heterocysts were found by the writer in the South Indian material of *N. lobatus* as was observed by Geitler in the Sunda Island material and by Frémy and Feldmann in the Banyuls material. As regards the third feature, viz., the solid nature of the thallus, it may be mentioned that the maximum size reached by the thallus is 1 cm. and the thallus was observed by the writer to be quite solid in all the stages. On the other hand the thalli of *N. lobatus* become hollow even before they reach a diameter of half a centimeter. Prof. Iyengar, who collected the alga, also states that the colonies of *N. radians* were quite solid throughout in all their stages and were never hollow.

The above three features of *N. radians* are quite distinct and do not bear any resemblance to those of *N. lobatus*. The writer, therefore, feels that *N. radians* should be retained as a distinct species, and is unable to accept Frémy and Feldmann's suggestion that *N. radians* should be included under *N. lobatus*.

One more form, viz., *N. lobatus* var. *stagnalis* Hansgirg (1893) still remains to be examined. According to Hansgirg (1893, p. 29), this variety has a solid thallus which is only 2-5 mm. in diameter and has filaments 4-6 μ broad. Geitler as well as Frémy and Feldmann doubt whether the alga is a *Nostochopsis* at all. Geitler (Geitler and Ruttner 1935-36) states that the figures given by Hansgirg (1893, figs. 5-6) (Text-figs. 34, 38) are not in total agreement with the general aspects of the genus *Nostochopsis*, since they lack the lateral heterocysts which are so characteristic of the genus. Frémy and Feldmann (1934) also expressed similar doubts. This question, as Frémy and Feldmann state, can be decided finally only when fresh material of this variety is again available for detailed examination. Even supposing that, on a further examination of fresh material of this variety, lateral heterocysts should be found in it and it should prove to be a *Nostochopsis* after all, since the dimensions of var. *stagnalis* as given by Hansgirg (1893, p. 29) fall within the range of the dimensions of *N. lobatus*, this variety, var. *stagnalis*, will have to be included under *N. lobatus* Wood emend. Geitler.

Before closing, a word may be said about the genus *Myxoderma* Schmidle. Schmidle (1900 c) first described this alga as a new species of *Nostochopsis*, which he called *N. Goetzei*, but later (1901) removed it from *Nostochopsis* and placed it in a new genus by name *Myxoderma*. This new genus is mainly characterised by a flat skin-like floating thallus with horizontal and erect filaments and lateral heterocysts (sessile and pedicellate). The horizontal filaments measure 2-3 μ



Text-figs. 29-33 and 35-37. *Nostochopsis Hansgirgii* Schmidle (after Schmidle, 1900b). Figs. 34 and 38. *Nostochopsis lobatus* var. *stagnalis* Hansgirg (after Hansgirg, 1893).

broad and the erect filaments about 5μ broad. This genus agrees in all respects with *Nostochopsis* except for the flat, skin-like nature of the thallus. Considerable doubt has been expressed regarding the validity of the genus by later workers. Borzi (1917, p. 139) includes this genus under *Nostochopsis*. Bharadwaja (1934, p. 6) states that "The retention of *Myxoderma*, which differs only in habit from *Nostochopsis*, also appears open to question". Frémy and Feldmann (1934, p. 615) also state that they are totally convinced that the genus *Myxoderma* represents only a stage in the life of *Nostochopsis* in which the thallus is still like a plate. The writer feels that the above

three authors are quite justified in doubting the validity of the genus. He thinks that the flat thallus of *Myxoderma* represents merely a fragment of the hollow thallus of *Nostochopsis lobatus* which was collected in a floating condition.

KEY TO SPECIES OF *Nostochopsis*

- I. Thallus solid when young and hollow when old
 - (1) Filaments 3-5.5 (-9) μ broad in the branching zone *N. lobatus*
- II. Thallus solid throughout
 - (1) Intercalary heterocysts present
 - (a) Filaments 1-3 μ broad in the branching zone *N. Hansgirgii*
 - (2) Intercalary heterocysts absent
 - (a) Filaments in the branching zone 4-8.4 μ broad *N. radians*

DESCRIPTION

Genus *NOSTOCHOPSIS* Wood

(*Mazaea* Bornet and Grunow and *Myxoderma* Schmidle)

(1) *Nostochopsis lobatus* Wood emend. Geitler*

(Including *N. Wichmannii* W. v. Bosse, *N. Hansgirgii* var. *sphaericus* Gardner and *Myxoderma Goetzei* Schmidle)

Thallus spherical or sub-spherical, more or less irregularly lobed, solid or hollow upto 3.5 μ diameter, seldom very big, blue-green or olive or yellowish-green. Trichomes radially arranged seemingly straight, in the innermost parts bent irregularly or in a zig-zag manner. The lower part of the upper region richly branched, in the older parts branches scattered; branches lateral, broadly stretching upwards by a short bend and with its end being parallel to the main filament. Cells in the branching zone barrel-shaped, 4-5.5 μ broad and just before the branch formation up to 9 μ broad, mostly up to 8 μ and in older regions up to 18 μ long. Trichome end tapering or slightly club-shaped in the other places usually 2 μ broad, apical cell distinctly pointed. Heterocysts mostly lateral (sessile) or terminal (pedicellate) on 2-3 or 4-celled lateral branch, more or less spherical or ellipsoidal, mostly 6-9 μ broad, seldom intercalary or basal (when young). Mucilage homogeneous, colourless or yellowish or brownish. Sheath present in the branching zone, thin and firm, seemingly colourless, or coloured yellowish to brownish, diffuent at the apices of branches, but persistent at the base. Hormogone on special hormogone branches, 3-3.5 μ broad and 8-20 cells length, cells short barrel-shaped.—In standing or in flowing waters, attached or free-floating.

* Adapted from Geitler (1933).

(2) *Nostochopsis Hansgirgii* Schmidle*

Thallus firm, spherical or irregular, yellowish-brown. Filaments sparsely branched, $1-2\mu$ broad, at the apex $0.6-0.9\mu$ broad, tapering to a rounded apex. Cells cylindrical, occasionally constricted at the joints, light blue-green, yellowish-brown or greyish-blue. Intercalary heterocysts long ellipsoidal or somewhat pointed at both ends, $4-5\mu$ broad and $6-10\mu$ long, lateral heterocysts spherical or hemispherical, 5μ broad.—On moist rocks.

(3) *Nostochopsis radians* Bharadwaja†

Thallus hemispherical or spherical, mucilaginous, blue-green, consisting of (1) a small compact base attached to the substratum and composed of densely arranged, irregularly curved and profusely branched filaments, (2) a middle region with loosely arranged straight or slightly curved, somewhat narrower, and less branched filaments running radially; and (3) an outer region of unbranched thread-like and uniformly broad filaments. Filaments in the older regions with a thin, hyaline and firm sheath. Branching true, irregularly arranged in the basal region, and generally unilateral in the middle region; when young perpendicular to the main filament, later bend and occupy a radial position. Trichome blue-green, cells in the basal region more or less rounded, sometimes barrel-shaped, deeply constricted at the cross-walls, up to 8.4μ broad; in the middle region elongated and more or less barrel-shaped in the inner parts and cylindrical with or without constriction towards the outer region, and up to 6.3μ broad; cells of the projecting threads elongate-cylindrical and $1.5-(2.1)-2.5\mu$ broad and $7-50\mu$ long. Heterocysts only lateral, sessile or pedicellate on 1-2 or 3-celled stalk, occurring in the basal and the middle region only; spherical, ovate or obovate; $4-9\mu$ broad and $6-15\mu$ long; intercalary heterocysts absent.—On submerged rocks in a stream.

Forma incertae sedis vel non satis descripta(4) *Nostochopsis lobatus* var. *stagnalis* Hansgirg‡

Thallus firm, spherical or irregularly shaped, mostly 2-5 mm. in diameter. Main filaments richly branched, radially disposed. Branches single, 2-3 near one another. Cells of the main filament adpressed, spherical or barrel-shaped, seldom long-ellipsoidal, mostly $4-6\mu$ broad, 1-2 times as long as broad. Heterocysts ellipsoidal or long-cylindrical, on the branches mostly $15-18\mu$ long, the rest as in the type form.—In standing water.

SUMMARY

Two species of the genus *Nostochopsis* Wood, viz., *N. lobatus* Wood emend. Geitler and *N. radians* Bharadwaja, collected from

* Adapted from Lemmermann (1910).

† Adapted from Bharadwaja (1934).

‡ Adapted from Hansgirg (1893).

South India are described in detail. An account is also given of the formation of the hormogones and all the stages of their development into young filaments in *N. lobatus*.

Geitler included *N. Wichmannii* W. v. Bosse and *N. Hansgirgii* var. *sphaericus* Gardner, under *N. lobatus* Wood emend. Geitler. Frémy and Feldmann approve of this change. The writer agrees with them regarding the inclusion of these forms under *N. lobatus* Wood emend. Geitler.

Frémy and Feldmann suggest that *N. Hansgirgii* Schmidle and *N. radians* Bharadwaja also should be included under it. The writer feels that there is no justification for including these two species under *N. lobatus*. He considers that *N. Hansgirgii* and *N. radians* are quite distinct from *N. lobatus*.

In conclusion, the writer wishes to express his great indebtedness and gratitude to Prof. M. O. P. Iyengar for his kind help and guidance throughout the investigation and to the authorities of the University of Madras for awarding a scholarship during the tenure of which the present investigation was carried out.

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